

Matthew Phillips

Vehicle Software + Full-Stack Product Engineer | Embedded-Adjacent Systems | AI Workflows | Operations Tools
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SUMMARY

Senior product engineer with a full-stack, vehicle-software, and hardware-connected background. I build real systems for real users: vehicle telemetry tools, BLE hardware apps, CAN bus decoding, firmware update workflows, AI support systems, ecommerce/admin platforms, CRM/order workflows, warranty and claims tools, dealer portals, dashboards, and internal automation.

My strongest work sits at the intersection of software and the physical world. I have built around JB4 tuning hardware, CANFlex fuel/sensor controllers, TurboLamik AWD/transmission telemetry, Stealth Batteries commerce/admin operations, Stealth Machine support tooling, customer tracking systems, AI agents, onboarding, warranties, shipping, and support workflows. I am comfortable taking an unclear problem, learning the domain, finding the real constraints, shipping a useful system, and improving it from feedback.

CORE STRENGTHS

- Vehicle and hardware-connected software: CAN bus workflows, BLE device communication, ECU/tuning workflows, telemetry parsing, firmware update flows, bootloader/erase/write/checksum patterns, diagnostics, datalogging, live gauges, and configuration tooling.
- Embedded-adjacent systems: ESP-IDF-oriented C scaffolding, CAN RX integration points, signal decoding, health/watchdog state, raw frame capture, derived metrics, shadow-mode control logic, and validation-first hardware integration.
- Full-stack product engineering: Node.js, Express, React, Next.js, TypeScript, JavaScript, Python, SQL/Postgres, SQLite, MongoDB/Mongoose, Supabase, Payload CMS, REST APIs, WebSockets, JWT auth, role permissions, and audit/logging patterns.
- AI and automation: Claude/Anthropic integrations, AI support agents, knowledge-base context, document ingestion, ticket triage, claims/warranty handling, onboarding workflows, internal workflow assistance, analytics, and human handoff paths.
- Operations software: CRM tools, order management, customer portals, dealer workflows, affiliate/sales-rep workflows, approval systems, shipping workflows, payment/webhook flows, dashboards, notifications, and support tooling.
- Working style: scope the problem, understand the user, keep changes readable, document decisions, communicate early, and own production behavior.

PROFESSIONAL EXPERIENCE

Freelance Full-Stack Product Engineer / Integration Developer

Stealth Batteries, Stealth Machine, AI Support Systems, Commerce/Admin, CRM, Operations Tools

Current

- Built full-stack applications and integrations for Stealth Batteries, Stealth Machine, and related businesses, owning ambiguous product problems from discovery through data modeling, frontend/backend development, debugging, deployment, and follow-up iteration.
- Built Stealth Batteries commerce/admin platform using Next.js, React, TypeScript, Payload CMS, Postgres, Stripe, webhooks, Resend/Nodemailer, Vercel Blob, UPS/shipping APIs, dealer portals, affiliate workflows, sales-rep workflows, order management, warranty/support workflows, and admin dashboards.
- Built Stealth Machine / Stealth Batteries AI support tooling with customer chat, onboarding, claim/warranty intake, internal workflow assistance, knowledge-base context, logging, analytics, ticketing, and human handoff paths.
- Built AI agent workflows that support customer service and internal operations across onboarding, warranties, shipping, support triage, product questions, document/context lookup, and repeatable employee-level workflows.
- Built CRM/order/customer tracking systems with Node/Express, SQLite, JWT auth, role permissions, admin approval workflows, customer portals, OCR/document scanning, real-time updates, Twilio/Nodemailer notifications, audit trails, and deployment documentation.
- Built automation and dashboards that connected customer-facing portals, admin work, order state, approvals, document uploads, notifications, reporting, and support history into usable business workflows.
- Used AI development tools daily for implementation, code review, documentation, refactoring, and debugging while retaining responsibility for correctness, security, and production behavior.

Burger Motorsports

Automotive Tuning, JB4Pro, CANFlex, Firmware-Adjacent Integration, Vehicle Software, BLE, CAN Bus

2019 onward

- Helped build and launch software around the JB4 tuning device ecosystem, including mobile application features, device-facing workflows, firmware-related flows, customer tools, diagnostics, support systems, and platform-specific behavior across many vehicle platforms and firmware combinations.
- Built JB4Pro mobile/device software using Flutter/Dart with BLE scan/connect behavior, connection recovery, notification parsing, ASCII/byte-level command handling, live gauges, datalogging, settings screens, user-adjustment interfaces, and developer diagnostics.

- Implemented firmware update workflows for JB4Pro-style hardware, including firmware download/manual HEX loading, bootloader entry, erase/write commands, block flashing, checksum validation, bootloader exit, progress state, and user-facing safety warnings.
- Built and maintained command/protocol logic for JB4 hardware, including settings retrieval, settings save, logging start/pause, map selection, E85/flex fuel configuration, WMI/meth settings, boost/fuel tables, AFR/meth/fuel pressure telemetry, and developer packet inspection.
- Developed CANFlex-style hardware software for fuel and sensor control: BLE device communication, ethanol content, fuel temperature, fuel pressure, CANbus output configuration, analog output modes, pressure sensor modes, calibration controls, firmware/version display, live gauges, logs, and race-car tuning workflows.
- Built CRM-backed support tooling that replaced manual email-based log review with structured customer submissions, vehicle log intake, AI-assisted parsing, diagnostic/tuning guidance, support history, and escalation paths.
- Worked close to embedded systems, firmware, custom hardware, and vehicle communication protocols, translating raw device/vehicle data into reliable UI state and supportable diagnostic workflows.
- Operated in a domain where incorrect parsing, settings, or firmware/version assumptions can affect real vehicles, requiring careful validation, documentation, and ownership of edge cases.

Independent Vehicle / Hardware Software Projects

TurboLamik AWD Controller, CANFlex, JB4Pro, Vehicle Telemetry and Integration Tools

- Built a TurboLamik AWD controller project with ESP-IDF-oriented firmware scaffolding for BMW E90 + TurboLamik integration, including passive CAN listening on a shared 500 kbps bus, selected E90 CAN ID decoding, TurboLamik TCU frame decoding, normalized drivetrain/chassis signals, derived metrics, health/watchdog state, raw CAN capture, and shadow-mode AWD request computation.
- Reverse engineered TurboLamik ADX protocol behavior into a client-device spec, including 921600 baud serial assumptions, LOG1/LOG2 polling sequence, fixed 100-byte packet maps, gearbox torque, engine torque, input/output RPM, slip/clutch data, linear pressure, lockup, analog inputs, CAN torque fields, AWD activation, PWM output, steering angle, and error/safe-mode data.
- Built BMW E90 PT-CAN-style signal maps for throttle, engine RPM, wheel speeds, coolant/oil temperature, steering angle, brake proxy, handbrake, reverse, and validation notes separating confirmed decodes from provisional in-car validation items.
- Documented MaxxECU default CAN output integration for 11-bit IDs 0x520-0x542 at 500 kbit/s, including RPM, throttle, MAP, lambda, ignition, fuel duty, vehicle speed, slip, traction flags, brake/clutch flags, accelerations, oil pressure/temp, boost target, transmission temp, differential temp, and firmware-version-gated protocol differences.
- Designed the Flutter telemetry/logging app shell for vehicle-side diagnostics, including device status, live dashboard, raw frame monitor, decoded signal monitor, capture control, saved sessions/export, and vehicle profile display.

Bayer

Technical / Analytical Testing Software Background

2015-2018

- Worked in analytical/scientific environments after studying biochemistry and software engineering.
- Built and supported custom code and analytical workflows for regulated testing and scientific/technical operations.
- Gained experience with careful documentation, data integrity, regulated processes, quality expectations, and technical troubleshooting.

SELECTED PROJECT HIGHLIGHTS

Stealth Batteries Commerce/Admin Platform

Built a production business platform connecting ecommerce, products, dealers, affiliates, sales reps, orders, shipping, support, warranty, dashboards, and Payload/Postgres admin tooling. The system tied customer-facing workflows to internal operations so the business could move faster without relying on manual email/status tracking.

Stealth Machine / Stealth Batteries AI Support Tools

Built AI-assisted customer and internal support systems with chat, onboarding, knowledge-base context, warranty/claims handling, ticketing, analytics, logging, escalation, and human handoff. These were practical workflow tools designed to reduce manual support load, not demo chatbots.

JB4Pro Vehicle Tuning Platform

Built mobile and hardware-connected software for JB4Pro tuning hardware, including BLE device discovery/connection, command handling, live telemetry, logging, firmware update flows, settings save/retrieval, map/user adjustment screens, E85/flex fuel workflows, WMI/meth configuration, boost/fuel tables, AFR/fuel pressure telemetry, and developer diagnostics.

CANFlex Fuel and Sensor Controller

Built CANFlex software for a fuel injector / E85 sensor / fuel pressure control ecosystem with BLE communication, ethanol/fuel-temperature/fuel-pressure gauges, CANbus and analog output configuration, pressure sensor modes, calibration, logs, firmware/version handling, and ECU/JB4 integration paths for race-car tuning.

TurboLamik AWD / Transmission Telemetry

Built an AWD controller and telemetry foundation around BMW E90 + TurboLamik, including CAN RX scaffolding, E90 decoder, TurboLamik decoder, raw CAN capture, health/watchdog state, derived metrics, shadow-mode AWD request logic, BLE service stub, Flutter live dashboard, raw frame monitor, decoded signal monitor, and saved log/export workflows.

Customer Tracking CRM

Built CRM/order workflow software with admin approvals, customer portal views, uploaded documents, OCR-style intake, audit trails, real-time updates, notifications, and role-aware workflows for messy business operations.

GEO Command Center

Built AI/search visibility tooling with multi-tenant dashboards, Supabase/Postgres, Drizzle, Redis/BullMQ-style workers, FastAPI services, Claude/Perplexity-style connectors, content audits, citation tracking, report generation, and analytics workflows.

TECHNOLOGIES

Vehicle / Embedded / Hardware: C, ESP-IDF-oriented scaffolding, CAN bus, CAN-system workflows, BLE, OBD2-connected hardware, ECU/tuning workflows, BMW E90 signal decoding, TurboLamik TCU frame decoding, MaxxECU CAN output docs, raw CAN capture, health/watchdog state, derived metrics, shadow-mode control logic, diagnostics, datalogging, gauges, firmware update flows, bootloader workflows, HEX parsing, checksums.

Full Stack: Node.js, Express, React, Next.js, TypeScript, JavaScript, REST APIs, WebSockets, JWT, role permissions, Payload CMS, Postgres, SQLite, MongoDB/Mongoose, Supabase, Drizzle, Redis/BullMQ-style queues.

Mobile / UI: Flutter, Dart, Android concepts, live dashboards, charts/gauges, configuration UI, raw frame monitors, decoded signal monitors, settings screens, customer portals, admin dashboards.

AI / Integrations: Claude/Anthropic, AI support agents, prompt/context workflows, document ingestion, OCR/document processing, Stripe, shipping APIs, UPS workflows, email/SMS providers, Resend/Nodemailer, Twilio, webhooks, logging, analytics.

Automation / Ops: Python, PowerShell, Docker Compose, FastAPI, Git, deployment notes, support tooling, internal scripts, technical documentation.

EDUCATION

Biochemistry and Software Engineering coursework, 2012-2015 Studied biochemistry with a software engineering focus while preparing to build analytical testing and technical workflow software.

HOW I WORK

I am comfortable with “here is the problem, go figure it out” work. I ask scope questions early, especially around users, safety, diagnostics, edge cases, ownership, and production risk. I prefer simple architecture, clear data flow, validation before control output, and code that another engineer can safely change. I communicate before going dark and leave notes/change logs so the system is easier to understand later.